

LAKEFRONT WINDOW

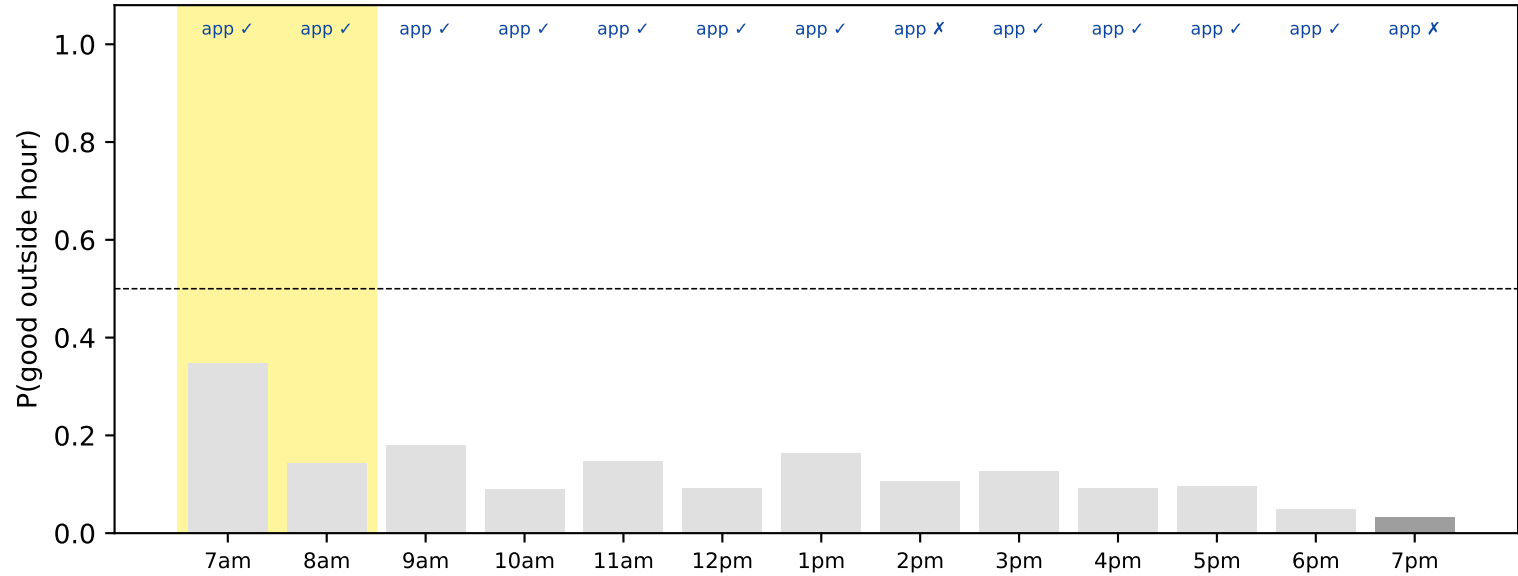
Built with PriorLabs-TabPFN

Oak Street Beach, Chicago · Thu Apr 18, 2024 · Replay of a holdout day

No great window. Least-bad: 7am-9am

Chance the lakefront is a 'good outside hour' in that window: 25% · sunrise 6:05 AM, sunset 7:34 PM

Hour by hour: TabPFN-v2 lakefront estimate (hatched = not full daylight)



Hour	Temp	Wind	Rain	Lakefront P(good)	App says OK?	Actual at lakefront
7am	50°F	5 mph	0%	35%	yes	good
8am	55°F	5 mph	0%	14%	yes	good
9am	56°F	6 mph	0%	18%	yes	good
10am	56°F	5 mph	0%	9%	yes	good
11am	56°F	10 mph	0%	15%	yes	good
12pm	57°F	9 mph	0%	9%	yes	good
1pm	57°F	9 mph	0%	16%	yes	not good
2pm	59°F	15 mph	100%	11%	no	good
3pm	59°F	10 mph	0%	13%	yes	good
4pm	59°F	9 mph	0%	9%	yes	good
5pm	58°F	5 mph	0%	10%	yes	good
6pm	56°F	10 mph	0%	5%	yes	good
7pm (dark)	53°F	14 mph	100%	3%	no	not good

What 'good' means: lakefront sensor reads 10-27 °C (50-81 °F), hourly gust under 9 m/s (~20 mph), no rain or snow.
How: TabPFN-v2 (open-weight tabular model, run locally on CPU) learned how the Oak Street lakefront sensor differs from Midway airport conditions (spring/summer lake cooling, onshore wind) and applies that to the inputs.
Holdout 2024-2026 (11,397 daylight-range hours, airport obs as input): TabPFN accuracy 91.4% vs 84.6% for 'trust the weather app as-is'; best-window hit rate 56% vs 51%. Real forecast error is NOT included in these numbers.
Inputs: actual Midway airport observations (a perfect 'forecast'); model trained on 2015-2023 only.
Data: Chicago Data Portal k7hf-8y75 · IEM ASOS archive (KMDW) · api.weather.gov. Not a safety tool: check beach and marine advisories. Built by Alfred, an AI agent, for Brooks Moore. MIT licensed.
Now fold this card, leave the phone, and go.